
Dynamic Evidential Routing: An Auditable Neurosymbolic Interface for Clinical Transformers Under Extreme Missingness

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Abstract

1 Dynamic evidential routing is an auditable inference-time interface for uncertainty-
2 conditioned symbolic intervention, not a demonstrated clinical performance im-
3 provement. The interface maps Monte Carlo dropout uncertainty into Non-
4 Axiomatic Logic (NAL) truth values, revises those values with explicit clinical
5 rules, and uses revised confidence to gate attention. Every intervention exports
6 its trigger, neural and symbolic truth values, revision, and attention effect. We
7 evaluate TCGA-THCA lymph node metastasis prediction as a clinical-genomic
8 feasibility check and WiDS ICU hospital-mortality prediction as the primary high-
9 missingness scale test. The symbolic path is active at scale, firing 13,031 times
10 across 8,551 of 13,757 held-out ICU stays. On WiDS, paired Brier and ECE differ-
11 ences favor NARS-gated routing over the ungated transformer, but comparisons
12 with flat-confidence and MC-confidence-only gating include zero. The results
13 therefore support confidence-gated, auditable intervention without isolating sym-
14 bolic revision as the cause of the calibration difference. The implementation is a
15 NAL truth-value interface, not a full Non-Axiomatic Reasoning System (NARS)
16 cognitive architecture or a clinically validated decision device.

17 1 Introduction

18 Clinical AI systems, including emerging generative and agentic systems, need an inspectable way to
19 represent weak evidence and expert intervention. A sparse ICU record may contain a few abnormal
20 measurements, many unmeasured variables, and an apparently precise risk score. Missingness,
21 irregular measurement, and cohort shift are common in hospital data, yet standard neural predictors
22 usually collapse the available evidence into one probability [Li et al., 2019b, Ovadia et al., 2019,
23 Ghassemi et al., 2020]. Any downstream assistant that consumes that score inherits its hidden
24 uncertainty, making an overconfident error difficult to defer, audit, or correct.

25 The challenge is that current model components expose different pieces of the problem but do not
26 connect them cleanly. Transformers are flexible function approximators for clinical prediction [Rasmy
27 et al., 2021, Azizi et al., 2021, Li et al., 2019a], but attention weights are not themselves evidential
28 confidences. Uncertainty estimators can summarize prediction instability, but they usually do not
29 provide a symbolic intervention point. Rule-based explanations can be readable, but post-hoc rules
30 often describe a prediction without changing the computation. What is missing is an inference-time
31 interface in which neural uncertainty, symbolic evidence, and attention control share an auditable
32 representation.

33 We propose dynamic evidential routing as that interface. A neural predictive distribution is first
34 mapped into a two-dimensional truth value (f, c) , where frequency f describes the balance of

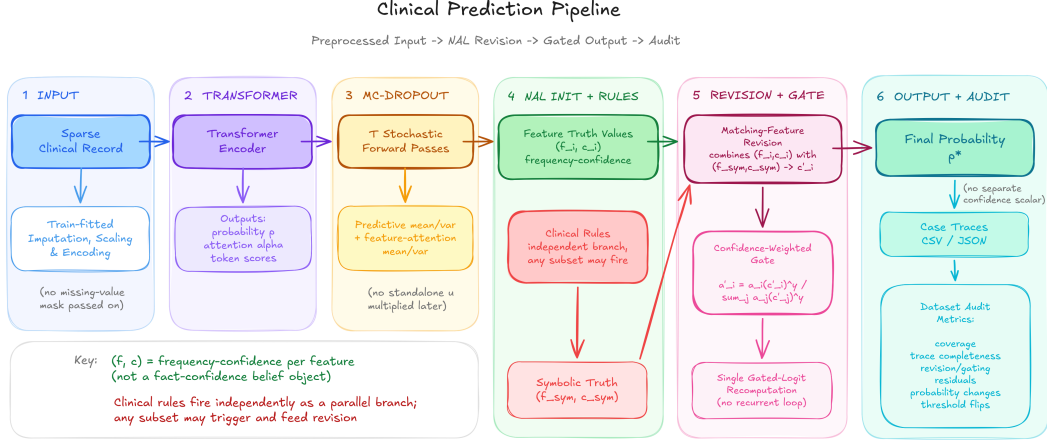


Figure 1: Inference-time dynamic evidential routing. A trained transformer produces a prediction and feature attention; MC dropout estimates uncertainty; the uncertainty signal initializes NAL truth values; triggered clinical rules revise those values; revised confidence gates attention; and the system exports a case-level trace showing where symbolic evidence changed the computation.

evidence and confidence c describes evidential support [Wang, 1995, 2006, Wang and Awan, 2013]. Explicit clinical rules then revise feature-level truths in the same NAL truth-value space, and the revised confidence values control a normalized attention gate. This design separates what the model predicts from how strongly the available evidence supports that prediction and exposes a trust signal that a clinical model, copilot, or agent can inspect before acting.

We evaluate the interface on two complementary clinical benchmarks. TCGA-THCA tests the mechanism under clinical-genomic feature fusion; WiDS ICU hospital-mortality prediction tests scale and extreme missingness. The contribution is an auditable inference-time interface: neural uncertainty and explicit rules share a truth-value representation, revised confidence can change attention, and the arithmetic is exported. The experiments do not establish a clinical performance improvement, and they do not isolate symbolic revision as the source of any calibration difference.

Figure 1 summarizes the inference-time data flow from sparse clinical inputs through uncertainty estimation, NAL-style truth initialization, rule-triggered revision, confidence-gated attention, and trace export.

2 Related Work

Neurosymbolic integration. Neurosymbolic AI combines neural pattern recognition with symbolic reasoning, knowledge injection, or interpretable rule systems [d’Avila Garcez et al., 2020, 2019, Bhuyan et al., 2024, Colelough and Regli, 2025]. Prior surveys distinguish loose coupling, tight coupling, and unified architectures [Hilario, 2013, Nawaz et al., 2025]. Dynamic evidential routing is a tight inference-time interface: the neural model supplies probabilities, uncertainty estimates, and attention weights, while the symbolic layer revises evidential confidence before attention is updated. Its contribution is the interface contract between these components. Neural uncertainty is converted into an explicit truth-value representation, symbolic rules revise that representation, and the revised confidence is fed back into attention with a trace of the intervention.

Clinical transformers and uncertainty. Clinical transformers have been used for structured EHRs, clinical text, and medical prediction tasks [Rasmy et al., 2021, Li et al., 2019a, Dai et al., 2025]. Attention can help interpretability, but medical attention mechanisms rarely represent uncertainty as a first-class control signal [Goncalves et al., 2022]. In parallel, uncertainty quantification methods such as Bayesian neural networks, MC dropout, and ensembles provide useful but imperfect epistemic uncertainty estimates [Blundell et al., 2015, Gal and Ghahramani, 2016, Lakshminarayanan et al., 2017, López et al., 2025]. MC dropout is attractive because it is cheap, but it can be miscalibrated under dataset shift [Ovadia et al., 2019]. Our use of uncertainty is therefore not a standalone calibration

method. The uncertainty estimate initializes evidential confidence, which becomes available for symbolic revision and case-level audit.

NARS and biomedical reasoning. NARS and Non-Axiomatic Logic model reasoning under insufficient knowledge and resources [Wang, 1995, 2006, 2022]. NAL truth values have been studied for uncertain medical diagnosis [Wang and Awan, 2013]. Biomedical knowledge graphs and symbolic reasoning can improve clinical interpretability [Wu et al., 2023, Abu-Salih et al., 2023, Rajabi and Kafaie, 2022, Jiang et al., 2024, Chudasama et al., 2025], but many such systems enrich representations rather than exposing an evidential confidence variable that directly controls neural inference. This paper uses NAL truth-value functions and rule-triggered revision only. It does not implement full NARS concept memory, task scheduling, resource allocation, or evidential-base management.

The closest families differ at the intervention boundary. Ordinary attention gating has no separately inspectable symbolic judgment; uncertainty calibration changes confidence without a rule-level intervention trace; post-hoc rules describe a completed decision; and training-time knowledge injection generally requires retraining to change the symbolic source. Dynamic evidential routing instead gives uncertainty and rules a shared truth-value interface, lets rules change inference in one pass, and preserves the source and arithmetic of that change. It does not claim a new general logic or a complete NARS architecture.

3 Method

3.1 Interface Boundary

The implementation should be read as a NAL truth-value interface inspired by NARS, not as a full NARS system. It uses the (f, c) truth-value representation, deduction-style grounding of triggered rules, and revision of neural and symbolic judgments. It does not maintain concept memory, task queues, priority dynamics, resource-bounded control, or explicit evidential-base objects. The claims in this paper are therefore interface-level claims about truth-value-guided fusion and attention control.

3.2 Truth Values

For a statement S with positive evidence w^+ and negative evidence w^- , NAL represents truth as (f, c) :

$$f = \frac{w^+}{w^+ + w^-}, \quad (1)$$

$$c = \frac{w^+ + w^-}{w^+ + w^- + k}. \quad (2)$$

We use the standard default $k = 1$, so $c = W/(W + 1)$ for total evidence $W = w^+ + w^-$. Strong deduction combines two premises by

$$f = f_1 f_2, \quad c = f_1 c_1 f_2 c_2. \quad (3)$$

Revision merges two judgments with weights $\omega_i = c_i/(1 - c_i)$:

$$f = \frac{\omega_1 f_1 + \omega_2 f_2}{\omega_1 + \omega_2}, \quad c = \frac{\omega_1 + \omega_2}{\omega_1 + \omega_2 + 1}. \quad (4)$$

Standard NAL revision assumes disjoint evidential bases [Wang, 2024]. The implementation treats neural and symbolic sources as distinct when revision is invoked, but it does not store evidential bases as runtime objects.

3.3 Neural-to-Symbolic Initialization

For a neural probability p and MC-dropout predictive variance σ^2 , the initial truth value is

$$f_{\text{neural}} = p, \quad c_{\text{neural}} = \frac{p(1 - p)}{p(1 - p) + \sigma^2 + \varepsilon}. \quad (5)$$

104 This is an application-specific initializer, not a canonical NARS measurement of evidential support.
 105 It parallels a variance-controlled evidence mass, then lets symbolic rules revise the confidence in
 106 NAL space. If the variance estimator underestimates uncertainty, c can be inflated. This limitation is
 107 inherited from the neural uncertainty method.

108 3.4 Symbolic Rule Revision

109 Each rule is a clinical proposition with a deterministic condition and a pre-specified symbolic truth
 110 value. A rule “fires” exactly when its condition evaluates true for the case. Firing does not depend
 111 on $(f_{\text{sym}}, c_{\text{sym}})$: f_{sym} encodes the prototype balance of supporting evidence and c_{sym} controls its
 112 evidential weight during revision. Triggered rules independently revise their matching feature truths
 113 by Eq. (4); any subset of rules may fire, and no conjunction of all four rules is required. The thresholds
 114 are interface probes, not learned decision thresholds or validated clinical cut points.

115 The symbolic truth values are prototype prior strengths, not an authoritative clinical knowledge base
 116 or literal outcome probabilities. A fired rule produces an inspectable event containing its identifier,
 117 condition, source feature, neural truth, symbolic truth, revised truth, and attention effect. A domain
 118 expert can therefore locate where a prior entered the model and can edit or disable that prior without
 119 retraining the transformer.

120 3.5 Attention Gating

121 Let α_i be the original attention weight for feature i and c_i the revised confidence associated with that
 122 feature. The gated attention is

$$\alpha_i^{\text{new}} = \frac{\alpha_i c_i^\gamma}{\sum_j \alpha_j c_j^\gamma}, \quad (6)$$

123 where γ controls the strength of confidence gating. The update preserves normalization and sets
 124 a feature’s contribution to zero when its confidence is zero. In this paper, $\gamma = 2$ is the default; a
 125 sensitivity run over $\gamma \in \{0.25, 0.5, 1, 2, 4\}$ is reported for TCGA.

126 3.6 Inference-Time Data Flow

127 The inference path has six stages. First, the trained transformer produces a point prediction, feature
 128 attention, and token scores. Second, repeated MC-dropout passes estimate predictive and attention
 129 variance. Third, the variance-to-confidence initializer in Eq. (5) creates feature-level neural truths.
 130 Fourth, the symbolic extractor emits every proposition whose condition is true. Fifth, neural and
 131 symbolic truths are revised in the shared (f, c) space. Sixth, revised confidence gates attention by
 132 Eq. (6); the model recomputes the final logit once and exports a trace. There is no recurrent feedback
 133 or convergence loop.

134 This staged design keeps the symbolic mechanism outside model training. With N cases, R rule
 135 predicates, and F features, predicate evaluation and gating cost $O(NR + NF)$ in the present
 136 vectorized implementation. The prototype assigns at most one rule to a logical feature and rejects
 137 collisions, so rule ordering cannot silently change a truth value. Larger bases need provenance-aware
 138 conflict policies, coverage analysis, and explicit aggregation of compatible or contradictory evidence
 139 before same-feature multi-rule revision can be supported.

140 3.7 Rule-Trace Semantics

141 For every triggered rule, the implementation records the feature name, rule condition, neural truth
 142 value, symbolic truth value, revised truth value, and resulting attention gain. These traces are not
 143 merely explanations after the fact. They correspond to actual arithmetic used by the prediction
 144 pipeline. A rule trace can therefore support three kinds of audit: whether the right rule fired, whether
 145 its symbolic truth value is clinically defensible, and whether the revised confidence changed attention
 146 in the expected direction.

147 This is the main neurosymbolic distinction from a post-hoc explanation. In a post-hoc explanation,
 148 symbolic language may describe a neural decision without changing the computation. In dynamic
 149 evidential routing, symbolic truth values become part of the computation itself. The trace can still be
 150 wrong if the rule base is wrong or the proposition extractor is misaligned, but the failure is localizable.

3.8 Worked Trace

For a neural estimate $p = 0.8$, $\sigma^2 = 0.05$, and a fired prototype rule $(f_{\text{sym}}, c_{\text{sym}}) = (0.9, 0.7)$, Eq. (5) gives $(f_{\text{neural}}, c_{\text{neural}}) \approx (0.8000, 0.7619)$. Revision gives $(f_{\text{rev}}, c_{\text{rev}}) \approx (0.8422, 0.8469)$. In natural language, the rule contributes relatively strong positive evidence with moderate confidence; it does not mean a clinician assigned a 0.9 patient-level risk.

4 Experimental Setup

We evaluate two benchmarks with different roles. TCGA-THCA predicts lymph-node metastasis from clinical and somatic-mutation features and tests whether the interface remains operational under clinical-genomic fusion. The labeled cohort contains 457 cases split into 319 training, 69 validation, and 69 held-out test cases. The processed input has 105 dimensions: 5 scaled clinical numeric features, 50 binary mutation indicators, and 50 one-hot clinical categorical features.

WiDS Datathon 2020 predicts in-hospital mortality for ICU encounters and is the primary validation under scale and extreme missingness. The run uses 91,713 labeled stays split into 64,199 training, 13,757 validation, and 13,757 held-out test stays, with 16 model input features after preprocessing.

All transformer variants use the same trained base architecture: $d_{\text{model}} = 64$, four attention heads, two encoder layers, and 50 MC-dropout passes for uncertainty estimation. We compare a random forest, a baseline transformer, flat-confidence gating with $c = 0.5$, MC-confidence-only gating, and NARS-gated symbolic revision. Metrics are AUC, Brier score, expected calibration error (ECE), and accuracy. Bootstrap confidence intervals are reported for AUC, Brier, and ECE.

4.1 Baselines and Controls

The baseline transformer tests whether the underlying neural architecture is competitive before confidence gating is introduced. The flat-confidence control assigns the same confidence to every feature and therefore isolates the architectural effect of adding a gate without using uncertainty. The MC-confidence-only ablation uses the neural variance-derived confidence but does not apply symbolic revision. The NARS-gated model adds the symbolic rule path on top of the MC-confidence initializer. This ordering is important: a difference from the baseline alone is not enough to show that symbolic reasoning helped, because confidence gating or regularization may explain the change.

The random forest baseline addresses the common tabular-learning concern that tree ensembles can be strong competitors on structured clinical features. It is trained on the same split and post-preprocessing design matrix, so differences cannot be attributed to a different data representation.

4.2 Symbolic Isolation, Uncertainty, and Shift

Tables 2 and 3 do not isolate symbolic revision. The evaluation contract now rescores NAL revision after four changes to the symbolic input: removing every rule, removing one rule at a time, permuting rule assignments across patients, replacing symbolic truth values with random draws, and substituting fixed symbolic priors. Rule removal must reproduce MC-confidence-only gating. Sensitivity scales symbolic confidence and recomputes revision; it does not retune the transformer on test labels. A claim that the symbolic component improves prediction would require the stated rules to separate from MC-confidence-only gating and from these controls. That separation is not a result of the saved runs.

The saved runs still initialize uncertainty with single-seed MC dropout. The implementation can also train a deep ensemble over seeds and compare MC-dropout variance, Bernoulli predictive entropy, and ensemble variance, including how each score ranks threshold errors. Those comparisons are not in the tables.

Each rule now carries a source note, a provenance identifier, an expert-review status, and a contradiction group. The current predicates are prototype probes and are marked not reviewed. Two rules that share a contradiction group and a target feature are rejected rather than overwritten. No independent cohort is evaluated here. A distribution-shift check can mask encoded features while holding the trained model, preprocessing, rules, gate, and decision threshold fixed. That check is a compatibility test under shift, not external clinical validation.

Table 1: Prototype symbolic rules. Conditions are evaluated independently; any subset may fire. Truth values are evidential weights for the interface, not validated clinical probabilities.

Data	Rule	Trigger condition	(f, c)	Clinical reading
TCGA	BRAF mutation	BRAF mutation indicator = 1	(0.85, 0.75)	The rule increases support for the BRAF feature when the mutation is recorded; it is not a validated risk probability.
TCGA	Age ≥ 55	age at diagnosis ≥ 55 years	(0.70, 0.60)	The rule raises evidential support for the age feature without asserting that age alone determines metastasis.
TCGA	Pathologic T3/T4	pathologic T category starts with T3 or T4	(0.90, 0.85)	The rule increases support for the observed staging feature; it does not establish nodal spread by itself.
TCGA	Extrathyroid extension	extrathyroid extension is Minimal (T3), Moderate/Advanced (T4a), or Very Advanced (T4b)	(0.85, 0.80)	The rule raises support only for an explicitly recorded extension category; missing, placeholder, and unknown values do not fire.
WiDS	Lactate	maximum day-1 lactate ≥ 4.0 mmol/L	(0.85, 0.80)	The rule increases support for the lactate feature without diagnosing shock or determining mortality.
WiDS	Hypotension	minimum day-1 systolic blood pressure ≤ 90 mmHg	(0.75, 0.70)	The rule increases support for the blood-pressure feature; it is not a stand-alone mortality decision.
WiDS	Age	age ≥ 75 years	(0.65, 0.60)	The rule increases support for the age feature without treating age as sufficient for mortality.
WiDS	Creatinine	maximum day-1 creatinine ≥ 2.0 mg/dL	(0.70, 0.65)	The rule increases support for the creatinine feature without diagnosing kidney injury or determining mortality.

4.3 Rule Bases

The TCGA rule base contains four thyroid-cancer rules chosen to exercise distinct input channels: BRAF mutation for genomic evidence, age at least 55 years for demographic evidence, pathologic T3/T4 stage for staging evidence, and extrathyroid extension for pathology-derived evidence. The WiDS rule base contains four ICU rules chosen to exercise common tabular critical-care signals: lactate for metabolic stress, hypotension for hemodynamic instability, age for demographic risk, and creatinine for renal dysfunction. All rules are applied only at inference time.

The small rule bases are prototype probes, not complete clinical ontologies. Clinical use would require expert-curated propositions, population-specific threshold rationales, provenance, and sensitivity analysis. Existing guideline or score components could be encoded and run in shadow mode to measure coverage, disagreement with extracted data, and downstream model effects. Such experiments could reveal where a rule interacts with the model; they would not validate the clinical correctness of the rule itself. Symbolic revision would be empirically isolated only if it separated from baseline, flat-confidence, and MC-confidence controls in paired comparisons. The current results do not meet that standard.

4.4 Statistical Reporting

The metric tables report point estimates followed by 95% bootstrap confidence intervals over held-out examples for AUC, Brier score, and ECE. Accuracy is reported as a point estimate. These intervals are confidence intervals, not standard deviations. This matters because the transformer variants are extremely close numerically. A table entry that is best at the fourth or fifth decimal place should not be interpreted as a confirmed improvement unless a corresponding interval or paired comparison excludes zero.

The paper therefore separates three evidential claims. First, whether the architecture compiles and runs on the benchmark. Second, whether the symbolic pathway fires and changes feature-level truth values. Third, whether the symbolic pathway changes predictive metrics beyond simpler controls. The present results support the first two claims. They do not isolate the third claim. This separation is central to the paper’s framing.

4.5 Reproducibility Interface

The public repository exposes a single endpoint for running both dataset branches and producing dataset-scoped metrics, traces, plots, and an aggregate run summary. The paper text reports the split sizes, input dimensions, architecture, MC-dropout count, and baselines needed to interpret the results. The code is still necessary for exact reproduction because preprocessing, imputation, categorical encoding, and rule extraction are implementation-sensitive.

The most important reproducibility constraint is that preprocessing must be fitted on the training split before being applied to validation and test examples. This prevents leakage from held-out data into

Table 2: Held-out TCGA-THCA performance for lymph node metastasis prediction. Point estimates are followed by 95% bootstrap confidence intervals where shown.

Variant	AUC	Brier	ECE	Acc.
Random forest	0.6613 [0.5101, 0.7908]	0.2200 [0.1895, 0.2498]	0.1597 [0.1317, 0.2824]	0.5942
Baseline transformer	0.7311 [0.5975, 0.8462]	0.2113 [0.1825, 0.2426]	0.1184 [0.1074, 0.2517]	0.6812
Flat-confidence	0.7328 [0.6028, 0.8479]	0.2109 [0.1816, 0.2427]	0.1178 [0.1130, 0.2641]	0.6812
MC-confidence-only	0.7311 [0.6008, 0.8453]	0.2110 [0.1820, 0.2418]	0.1184 [0.1041, 0.2481]	0.6812
NARS-gated	0.7311 [0.6008, 0.8453]	0.2109 [0.1820, 0.2417]	0.1184 [0.1042, 0.2481]	0.6812

Table 3: Held-out WiDS ICU performance for hospital mortality prediction. Point estimates are followed by 95% bootstrap confidence intervals where shown.

Variant	AUC	Brier	ECE	Acc.
Random forest	0.8768 [0.8669, 0.8868]	0.05817 [0.05520, 0.06142]	0.00737 [0.00596, 0.01254]	0.9258
Baseline transformer	0.8803 [0.8700, 0.8902]	0.05653 [0.05371, 0.05966]	0.00750 [0.00493, 0.01211]	0.9285
Flat-confidence	0.8802 [0.8698, 0.8901]	0.05648 [0.05366, 0.05964]	0.00618 [0.00421, 0.01098]	0.9286
MC-confidence-only	0.8802 [0.8698, 0.8901]	0.05647 [0.05365, 0.05962]	0.00581 [0.00409, 0.01062]	0.9286
NARS-gated	0.8802 [0.8698, 0.8901]	0.05647 [0.05365, 0.05962]	0.00583 [0.00411, 0.01064]	0.9286

imputation or encoding. The rule extractor is also run only at inference time, so symbolic rules do not alter the supervised training objective.

5 Results

The central positive result is operational: symbolic interventions execute on both modalities, expose their arithmetic, and retain predictive behavior close to the ungated transformer and confidence-only controls. Performance tables are reported to establish compatibility, not superiority.

On TCGA-THCA, the transformer variants remain tightly clustered. Flat-confidence has the highest AUC at 0.7328 and the lowest Brier point estimate at the displayed precision; the 69-case test split does not statistically separate variants. TCGA is therefore a multimodal feasibility test rather than evidence of symbolic-gating superiority.

The TCGA symbolic rules were active under clinical-genomic fusion. Counts and coverage below use only recognized extrathyroid-extension categories; placeholder and unknown values cannot fire.

WiDS provides the clearer scale test. All transformer variants have higher AUC point estimates than the random forest in this run, although their AUC confidence intervals overlap. Within the transformer family, the baseline has the highest AUC, while MC-confidence-only has the numerically lowest Brier and ECE point estimates. NARS-gated is nearly identical to MC-confidence-only and ties it and flat-confidence on accuracy at the reported precision.

Figure 2 visualizes the Brier-score intervals for both benchmarks. The overlap among transformer variants is large, especially on TCGA, reinforcing the interpretation that the current evidence supports operational compatibility rather than a separated performance claim.

The symbolic pathway operated at scale: ICU rules fired on 8,551 of 13,757 held-out stays, with 13,031 feature-level revisions. The per-rule counts were 1,936 for lactate, 5,338 for hypotension, 3,443 for age, and 2,314 for creatinine. Every fired rule mapped to a feature, produced a finite trace, and changed confidence and attention. The corresponding TCGA rates were also 100% across 79 events in 42 of 69 cases. Independently recomputed revision and normalized-gating residuals were zero for both datasets.

Table 4 evaluates operational auditability over every held-out rule event. Mapping coverage asks whether a fired proposition reached a model feature; completeness asks whether all values needed for inspection are finite; and residuals independently recompute NAL revision and normalized gating. Among triggered cases, the mean absolute baseline-to-NARS probability change was 0.00205 on TCGA and 0.00214 on WiDS, with maxima of 0.00600 and 0.03102. No triggered TCGA case crossed the 0.5 threshold, while 10 triggered WiDS cases did. These are system-fidelity checks, not evidence that clinicians find the traces useful or that a changed decision is clinically correct.

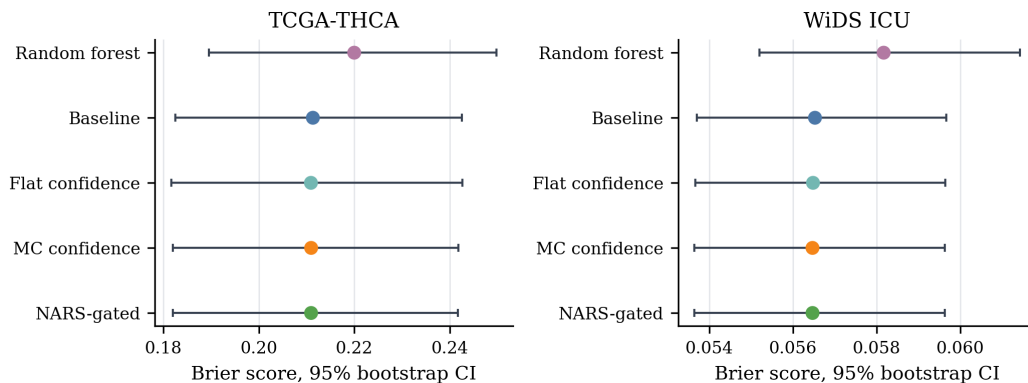


Figure 2: Held-out Brier scores with 95% bootstrap confidence intervals. Lower values are better. The transformer variants are close, so small point-estimate differences should be interpreted as descriptive unless corroborated by additional comparisons.

Table 4: Operational auditability on held-out cases. Completeness and residuals test whether traces faithfully expose the implemented arithmetic; they do not measure clinician usability or rule validity.

Data	Cases covered	Events	Complete	<i>c</i> changed	Flips	Max residual
TCGA-THCA	42/69 (60.9%)	79	100.0%	100.0%	0	0.0e+00
WiDS ICU	8,551/13,757 (62.2%)	13,031	100.0%	100.0%	10	0.0e+00

5.1 Case-Level Rule Trace

The refreshed prediction export provides a concrete held-out example. For WiDS encounter 8205 (target 0), the ungated probability was 0.5006, flat-confidence gating produced 0.4995, MC-confidence-only gating produced 0.4961, and NARS-gated routing produced 0.4968. Three rules fired: hypotension, lactate, and creatinine. Their attention weights changed from 0.2564 to 0.2526, 0.0231 to 0.0248, and 0.0347 to 0.0372, respectively. The trace therefore exposes a real threshold change and the rule-level arithmetic, but it also shows why attribution requires controls: MC-confidence-only had already crossed the threshold before symbolic revision. Across the complete WiDS export, MC-confidence-only and NARS-gated produced no different threshold decisions. This is operational auditability, not clinical validation.

5.2 Calibration Interpretation

The WiDS calibration results support a two-part interpretation. Relative to the ungated transformer, NARS-gated routing lowers Brier by 0.0000587 with a paired 95% interval of [0.0000249, 0.0000911], and lowers ECE by 0.00120 with an interval of [0.0000282, 0.00202]. Both intervals exclude zero. Relative to flat-confidence and MC-confidence-only gating, however, the paired Brier and ECE intervals include zero; MC-confidence-only also has the slightly better Brier and ECE point estimates. Thus the run supports a small benefit from confidence gating, but does not identify symbolic revision as its cause.

That restraint keeps the contribution aligned with what the experiment actually supports. Dynamic evidential routing should not be described as a universal calibration booster. The supported claim is that a symbolic evidential channel can be made active, traceable, and compatible with competitive calibrated prediction on a large high-missingness ICU benchmark.

6 Discussion and Ablation

The flat-confidence and MC-confidence-only controls are essential for interpreting the result. If NARS-gated attention were compared only with the baseline transformer, its paired Brier and ECE differences could be over-interpreted as a symbolic reasoning effect. The controls show a more precise picture: those differences are not separated from confidence gating before symbolic revision is added.

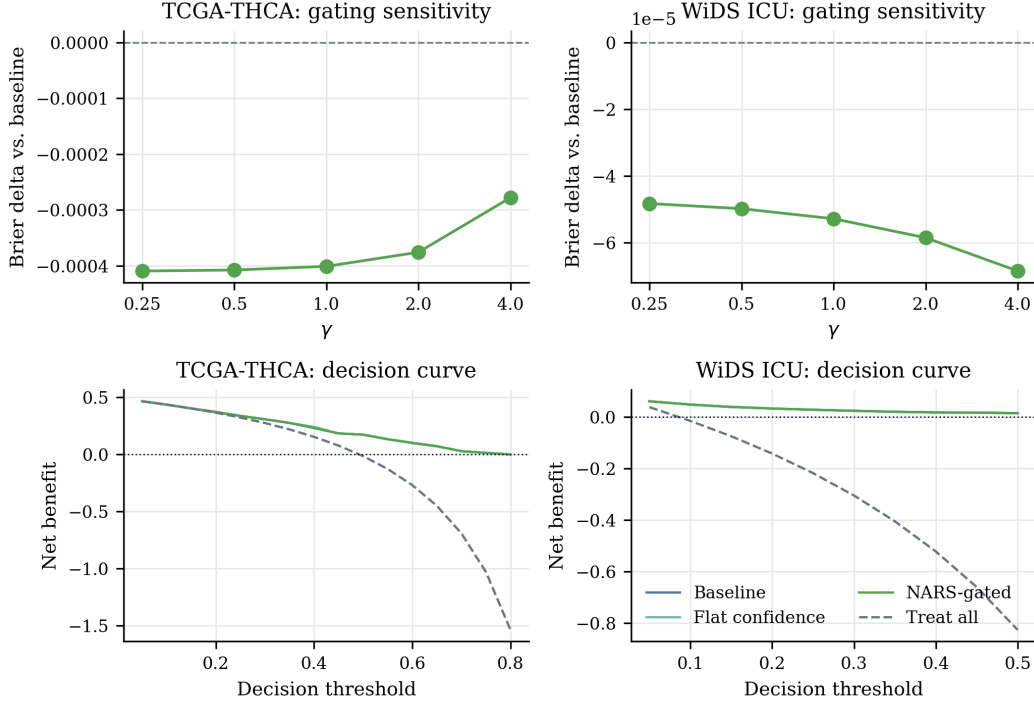


Figure 3: Reported ablations. Top: NARS-gated Brier change relative to the ungated transformer across γ (negative is lower). Bottom: held-out decision curves over informative threshold ranges. These descriptive curves do not establish clinical utility.

295 The NARS rule path is therefore best interpreted as an instrumentation layer whose demonstrated
 296 value is auditability, modifiability, and compatibility with symbolic intervention, not yet a statistically
 297 isolated source of within-family calibration change.

298 Figure 3 reports the gamma sweeps and decision curves rather than treating them as future experiments.
 299 Across $\gamma \in \{0.25, 0.5, 1, 2, 4\}$, Brier deltas remain small. On WiDS, stronger gating monotonically
 300 lowers the Brier point estimate relative to baseline, reaching a delta of -6.85×10^{-5} at $\gamma = 4$, while
 301 AUC deltas remain slightly negative. On TCGA, all five Brier deltas are negative but the small split
 302 gives no separated superiority regime. Decision curves likewise show closely overlapping transformer
 303 variants with modest threshold-dependent differences.

304 6.1 Failure Modes and Deployment Boundary

305 The interface can fail when MC dropout underestimates uncertainty, proposition extraction misaligns
 306 encoded features with clinical concepts, or rule truth values are weak or disputed. The trace localizes
 307 these failures: a reviewer can verify the extracted condition, inspect the rule and revised truth values,
 308 and rerun a case after editing or disabling the rule. This supports human oversight, but it is not a
 309 clinical safety mechanism.

310 The current system remains a methods prototype rather than a clinical decision device. Deployment
 311 would require a rule base curated independently by domain experts, external validation without
 312 retuning, shift monitoring, rule governance, human-factors evaluation, and clinician override. Stronger
 313 evidence for symbolic revision would also require separation from baseline, flat-confidence, and
 314 uncertainty-only controls across external cohorts.

315 7 Limitations

316 The current evidence is a methods evaluation, not a clinical validation study. TCGA has only 69
 317 held-out cases, WiDS uses an internal split rather than an external cohort, and both rule bases are

318 prototype probes that have not received expert review. Ensemble and frozen-shift comparisons are
 319 implemented, but they are not results in the reported tables. The neural-to-NAL initializer can also
 320 over-trust a prediction when MC dropout underestimates uncertainty under shift; future work should
 321 compare ensembles and shift-aware estimators on temporally or institutionally independent cohorts.

322 Finally, the repository implements NAL truth-value functions and feature-level rule revision, but not
 323 the complete NARS architecture. It does not maintain concept priority dynamics, task scheduling, or
 324 explicit evidential-base records. The claims should therefore be read as interface-level claims about
 325 truth-value-guided fusion and attention control.

326 8 Conclusion

327 Dynamic evidential routing is an auditable inference-time interface for uncertainty-conditioned
 328 symbolic intervention. Explicit rules can revise confidence and change attention, and the run leaves
 329 arithmetic traces. The saved comparisons do not show that symbolic revision improves clinical
 330 prediction. External cohorts, expert-curated rules, reported ensemble comparisons, and human
 331 evaluation are needed before clinical use.

332 **Ethics and reproducibility.** This study uses public, de-identified benchmarks and collected no
 333 new patient data. An anonymized implementation accompanies the submission. Large language
 334 models assisted with drafting, formulation, and code generation; the authors checked the derivations,
 335 implementation, references, and results and take responsibility for the work.

336 References

- 337 B. Abu-Salih et al. Healthcare knowledge graph construction: A systematic review of the state-of-
 338 the-art, open issues, and opportunities. *Journal of Big Data*, 10(1):83, 2023.
- 339 S. Azizi et al. Big self-supervised models advance medical image classification. In *Proceedings of*
 340 *the IEEE International Conference on Computer Vision*, pages 3478–3488, 2021.
- 341 Bikram Pratim Bhuyan, Amar Ramdane-Cherif, Ravi Tomar, and T. P. Singh. Neuro-symbolic
 342 artificial intelligence: A survey. *Neural Computing and Applications*, 36(21):12809–12844, 2024.
 343 doi: 10.1007/s00521-024-09960-z.
- 344 C. Blundell et al. Weight uncertainty in neural networks. In *Proceedings of the International*
 345 *Conference on Machine Learning*, pages 1613–1622, 2015.
- 346 Y. Chudasama et al. Towards interpretable hybrid AI: Integrating knowledge graphs and symbolic
 347 reasoning in medicine. *IEEE Access*, 13:12345–12360, 2025.
- 348 B. C. Colelough and W. Regli. Neuro-symbolic AI in 2024: A systematic review. arXiv:2501.05435,
 349 2025.
- 350 L. Dai et al. Automated classification of clinical diagnoses in electronic health records using
 351 transformer. *PLoS One*, 20(1):e0329963, 2025.
- 352 A. d’Avila Garcez et al. Neurosymbolic AI: The 3rd wave. *Artificial Intelligence*, 294:103504, 2020.
- 353 Artur d’Avila Garcez, Marco Gori, Luis C. Lamb, Luciano Serafini, Michael Spranger, and Son N.
 354 Tran. Neural-symbolic computing: An effective methodology for principled integration of ma-
 355 chine learning and reasoning. *Journal of Applied Logics – IfCoLog Journal of Logics and their*
 356 *Applications*, 6(4):611–631, 2019.
- 357 Y. Gal and Z. Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty
 358 in deep learning. In *Proceedings of the International Conference on Machine Learning*, pages
 359 1050–1059, 2016.
- 360 Marzyeh Ghassemi, Tristan Naumann, Peter Schulam, Andrew L. Beam, Irene Y. Chen, and Rajesh
 361 Ranganath. A review of challenges and opportunities in machine learning for health. *AMIA Joint*
 362 *Summits on Translational Science Proceedings*, 2020:191–200, 2020. URL <https://pmc.ncbi.nlm.nih.gov/articles/PMC7233077/>.

364 T. Goncalves et al. A survey on attention mechanisms for medical applications: Are we moving
365 toward better algorithms? *IEEE Access*, 10:98975–99003, 2022.

366 M. Hilario. An overview of strategies for neurosymbolic integration. In *Connectionist-Symbolic*
367 *Integration*, pages 15–34. Routledge, 2013.

368 P. Jiang et al. Reasoning-enhanced healthcare predictions with knowledge graph community retrieval.
369 arXiv:2410.04585, 2024.

370 B. Lakshminarayanan et al. Simple and scalable predictive uncertainty estimation using deep
371 ensembles. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

372 F. Li et al. Fine-tuning bidirectional encoder representations from transformers (BERT)-based models
373 on large-scale electronic health record notes: An empirical study. *JMIR Medical Informatics*, 7(3):
374 e14830, 2019a.

375 Xuedong Li, Yue Wang, Dongwu Wang, Walter Yuan, Dezhong Peng, and Qiaozhu Mei. Improving
376 rare disease classification using imperfect knowledge graph. *BMC Medical Informatics and*
377 *Decision Making*, 19(Suppl. 5):238, 2019b. doi: 10.1186/s12911-019-0938-1.

378 L. Julián Lechuga López, Shaza Elsharief, Dhiyaa Al Jorf, Firas Darwish, Congbo Ma, and Farah E.
379 Shamout. Uncertainty quantification for machine learning in healthcare: A survey, 2025. URL
380 <https://arxiv.org/abs/2505.02874>.

381 Uzma Nawaz, Mufti Anees ur Rahaman, and Zubair Saeed. A review of neuro-symbolic AI integrating
382 reasoning and learning for advanced cognitive systems. *Intelligent Systems with Applications*, 26:
383 200541, 2025. doi: 10.1016/j.iswa.2025.200541.

384 Y. Ovadia et al. Can you trust your model’s uncertainty? evaluating predictive uncertainty under
385 dataset shift. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

386 E. Rajabi and S. Kafaie. Knowledge graphs and explainable AI in healthcare. *Information*, 13(10):
387 459, 2022.

388 L. Rasmy et al. Med-BERT: Pretrained contextualized embeddings on large-scale structured electronic
389 health records for disease prediction. *npj Digital Medicine*, 4(1):86, 2021.

390 P. Wang. *Non-Axiomatic Reasoning System: Exploring the Essence of Intelligence*. PhD thesis,
391 Indiana University, 1995.

392 P. Wang. *Rigid Flexibility: The Logic of Intelligence*. Springer, 2006.

393 P. Wang. Non-axiomatic logic: A model of intelligent reasoning. Second edition draft manuscript
394 dated December 14, 2024, shared by author, 2024.

395 P. Wang and S. Awan. Reasoning in non-axiomatic logic: A case study in medical diagnosis. In
396 *Proceedings of the Conference on Artificial General Intelligence*, pages 185–190, 2013.

397 Pei Wang. Intelligence: From definition to design. In *Proceedings of the Third International*
398 *Workshop on Self-Supervised Learning*, volume 192 of *Proceedings of Machine Learning Research*,
399 pages 35–47. PMLR, 2022. URL <https://proceedings.mlr.press/v192/wang22a.html>.

400 X. Wu et al. Medical knowledge graph: Data sources, construction, reasoning, and applications. *Big*
401 *Data Mining and Analytics*, 6(2):201–217, 2023.